

GM Bus Tracking Platform — Documentation

Version: 1.0.0 | **Date:** Sept 2026 **Domain:** bustracker.gauravmedia.in | **IP:** 103.216.236.194 **Path:** /data/movies/Project/GM BUS STABLE

1. Overview

Multi-tenant SaaS for bus fleet management with real-time GPS tracking and multilingual voice announcements.

- Multi-tenant (separate DB per client)
- 4 languages: English, Marathi, Gujarati, Hindi
- Real-time GPS tracking
- Automated TTS announcements
- Admin + Client + Driver + Passenger apps

URLs

- Backend: <https://api.bustracker.gauravmedia.in>
 - Client: <https://app.bustracker.gauravmedia.in>
 - Admin: <https://admin.bustracker.gauravmedia.in>
 - Local: 5000 (backend), 5173 (admin), 5174 (client)
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2. Architecture

Admin Panel → Backend (Node.js + PostgreSQL) → Client Dashboard + Driver App ↓
ESP32 GPS Tracker

3. Tech Stack

- **Backend:** Node.js, Express, PostgreSQL, JWT, google-tts-api, PM2
 - **Frontend:** React 18 + TypeScript, Vite, Fluent UI, Leaflet, Recharts
 - **Mobile:** Flutter 3.x
 - **Infra:** Nginx Proxy Manager, Let's Encrypt, Cloudflare
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4. Implemented Features

Admin Panel: Login, Client List, Add Client (auto DB), Language Set, Toggle Active, Delete, Subscriptions, Analytics

Client Dashboard: KPIs, Live Map, Activity, Chart, Buses/Routes/Stops CRUD, Voice Settings, Drivers, Trip History, CSV Export, Dark Mode, 4-language UI, Toast, Search

Voice: Google TTS (4 langs), Rules (GU → GU+HI+EN), Cache 24h, Preview Player

5. Roadmap

- **Critical:** WebSocket, Driver App, Auto announcement, Geofencing
 - **High:** Email/SMS/Push, WhatsApp, PDF/Excel, Roles, 2FA
 - **Medium:** Maintenance log, Document alerts, Fuel tracking, Passenger app
 - **Low:** Multi-branch, White-label, Public API, Payment, GST
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6. Database

- **Master (gm_master):** clients
 - **Tenant (per client):** buses, routes, stops, live_locations, trips, settings
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7. API Endpoints

- **Admin:** /api/admin/login, /clients, /stats, /api/billing/*
 - **Client:** /api/auth/login, /api/bus, /api/route, /api/tracking/*, /api/trips
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8. Hardware — ESP32 Tracker

BOM (~Rs 1020-1550):

- ESP32 Dev Board: 340-450
- NEO-6M GPS: 230-350
- SIM800L GSM: 250-400
- LM2596 Buck: 100-150
- Wires/Case: 100-200

Wiring:

- ESP32 ↔ GPS: GPIO16(RX2) → TX, GPIO17(TX2) → RX, 3.3V, GND
- ESP32 ↔ SIM800L: GPIO4 → TX, GPIO2 → RX, 5V, GND

AIS-140: Commercial સાથી certified VLTD અનિવાર્ય.

9. Deployment

- **PM2:** 3 apps (backend 5000, client 5174, admin 5173)
- **Commands:** pm2 list / logs / restart all / monit / save
- **NPM Setup:** DNS 3 A records + 3 proxy hosts with SSL
- **Backup:** scripts/backup.sh (daily, 7-day retention)

10. Operations

Credentials:

- Admin: admin@gmbus.com / Admin@123
- Client: gaurav@test.com / Gaurav@123

Troubleshoot:

- Login fail → check backend/.env
 - Port in use → `kill -f "vite --port XXXX"`
 - PM2 down → `pm2 resurrect`
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11. Statistics

- Features Done: 60+
 - Features Planned: 70+
 - Languages: 4
 - API Endpoints: 30+
 - DB Tables: 8
 - PM2 Services: 3
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Contact

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